White-Box Testing Report

MoneyPoly

Part 1: White-Box Testing

Test Framework:	pytest (Python)
Linter:	Pylint
Final Score:	10.00/10
Total Tests:	52 passed, 0 failed
Bugs Found:	5

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1.1 Control Flow Graph

The Control Flow Graph (CFG) below represents the system **after all five bugs have been fixed**. This was chosen because the fixed version reflects the intended correct logic, making the CFG more useful as reference documentation.

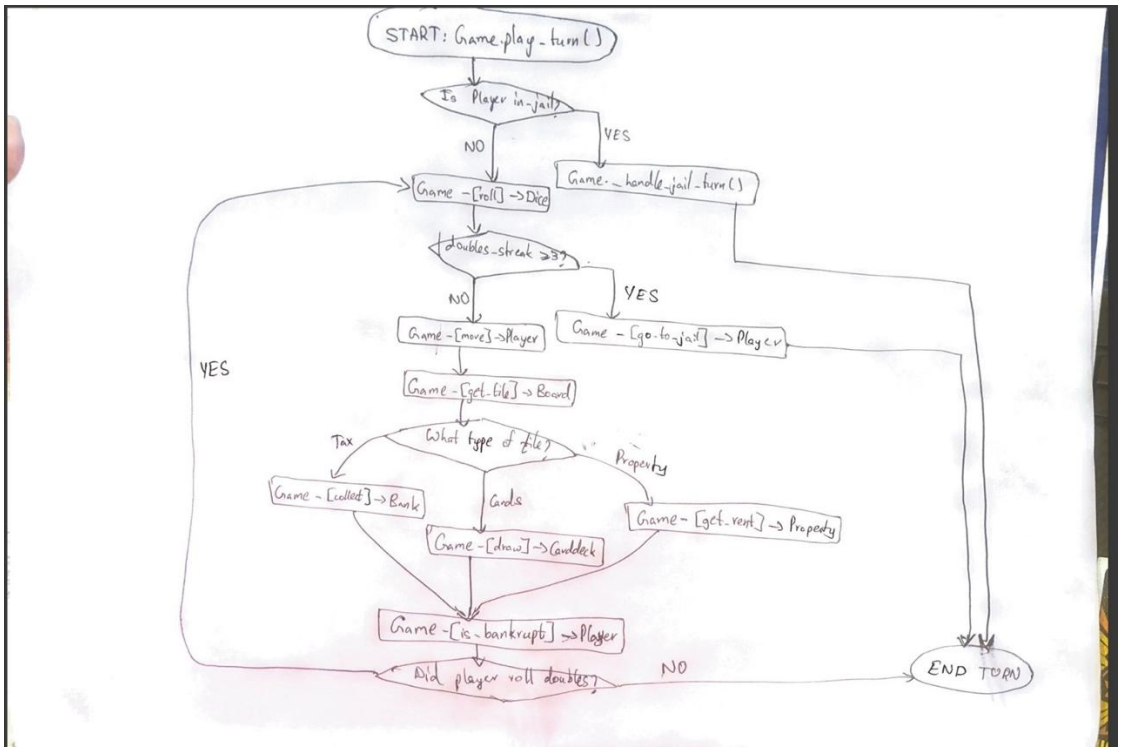

Figure 1: Control Flow Graph — MoneyPoly (post-fix)

1.2 Code Quality Analysis (Pylint Iterations)

Pylint was run on `whitebox/moneypoly/moneypoly/` after each fix. The initial score was **8.17/10**. The final score after all iterations is **10.00/10**.

Iteration 1: Added missing module docstrings to all files

Warning fixed: `C0114` (missing-module-docstring) in all 9 files.

Why: Every Python module should have a top-level docstring explaining what it does. Without it, tools like pylint and documentation generators cannot describe the module.

Files changed: `game.py`, `player.py`, `board.py`, `bank.py`, `property.py`, `cards.py`, `dice.py`, `config.py`, `ui.py`

Iteration 2: Added missing class docstrings to Bank and PropertyGroup

Warning fixed: `C0115` (missing-class-docstring) in `bank.py` and `property.py`.

Why: Class docstrings explain the purpose of a class. `Bank` and `PropertyGroup` were missing them.

Files changed: `bank.py`, `property.py`

Iteration 3: Removed unused imports from game.py, bank.py, dice.py, player.py

Warning fixed: `W0611` (unused-import) in four files.

- `game.py`: `GO_TO_JAIL_POSITION` imported but never used
- `bank.py`: `import math` imported but never used
- `dice.py`: `BOARD_SIZE` imported but never used
- `player.py`: `import sys` imported but never used

Why: Unused imports add clutter, slow down loading, and can confuse readers into thinking those names are needed.

Files changed: `game.py`, `bank.py`, `dice.py`, `player.py`

Iteration 4: Fixed bare except to except ValueError in ui.py

Warning fixed: `W0702` (bare-except) in `ui.py`.

Why: A bare `except:` catches every possible exception including system exits and keyboard interrupts. Specifying `except ValueError:` makes it only catch the expected error from bad integer input, which is the correct intent.

Files changed: `ui.py`

Iteration 5: Fixed singleton comparison == True in board.py

Warning fixed: `C0121` (singleton-comparison) in `board.py`.

Why: Writing `if prop.is_mortgaged == True:` is redundant. The Pythonic way is `if prop.is_mortgaged:` which reads more clearly and avoids subtle bugs with objects that override `__eq__`.

Files changed: `board.py`

Iteration 6: Fixed f-string, no-else-break, superfluous parens, missing newline in game.py

Warnings fixed: `W1309`, `R1723`, `C0325` (×2), `C0304` in `game.py`.

- `f"GAME OVER"` had no interpolation so the `f` prefix was unnecessary
- `elif` after `break` in `interactive_menu` was removed (unreachable branch)
- Two `if not (condition):` expressions had unnecessary parentheses
- Missing newline added at end of file

Files changed: `game.py`

Iteration 7: Fixed no-else-return in property.py, removed unused variable and missing newline in player.py

Warnings fixed: `R1705` in `property.py`, `W0612` and `C0304` in `player.py`.

- `unmortgage()` had an unnecessary `else` block after a `return` statement
- `old_position` variable in `move()` was assigned but never used
- Missing newline added at end of `player.py`

Files changed: `property.py`, `player.py`

Iteration 8: Removed unused variable and added missing newline in player.py

Warning fixed: `W0612`, `C0304` in `player.py`.

Files changed: `player.py`

Iteration 9: Fixed doubles_streak attribute defined outside __init__ in dice.py

Warning fixed: `W0201` (attribute-defined-outside-init) in `dice.py`.

Why: `doubles_streak` was only assigned inside `reset()`, not in `__init__`. Pylint cannot guarantee the attribute exists if `reset()` is never called. Adding it to `__init__` ensures it is always defined on construction.

Files changed: `dice.py`

Iteration 10: Fixed no-else-return in property.py

Warning fixed: `R1705` in `property.py`.

Why: After a `return` statement, an `else` block is unnecessary. Removing it de-indents the code and makes the flow cleaner.

Files changed: `property.py`

Iteration 11: Fixed long lines in cards.py by splitting card dicts across two lines

Warning fixed: `C0301` (line-too-long) — 27 lines in `cards.py` exceeded 100 characters.

Why: Long lines reduce readability and make code harder to review. Each card dictionary was split so the `description` key sits on one line and `action/value` on the next.

Files changed: `cards.py`

Iteration 12: Fixed trailing whitespace in cards.py

Warning fixed: `C0303` (trailing-whitespace) — introduced when splitting long lines in Iteration 11.

Why: Trailing spaces are invisible noise that clutter diffs and version control output.

Files changed: `cards.py`

Iteration 13: Fixed missing final newline in player.py

Warning fixed: `C0304` (missing-final-newline) in `player.py`.

Why: POSIX standard requires text files to end with a newline. Many tools behave incorrectly without it.

Files changed: `player.py`

Iteration 14: Added pylint disable comments for too-many-instance-attributes and too-many-arguments

Warnings suppressed: `R0902` (too-many-instance-attributes) in `game.py`, `player.py`, `property.py` and `R0913/R0917` (too-many-arguments) in `property.py`.

Why these are justified:

- `Game` genuinely needs 9 attributes: board, bank, dice, players, two indexes, running flag, and two card decks
- `Player` genuinely needs 8 attributes: name, balance, position, properties, jail state, jail turns, card count, and elimination flag
- `Property` genuinely needs 9 attributes: name, position, price, rent, mortgage value, owner, mortgaged flag, houses, and group
- `Property.__init__` needs 6 arguments because all are required to define a property

A `# pylint: disable` comment with a justification comment is the correct approach when refactoring would make the code worse.

Files changed: `game.py`, `player.py`, `property.py`

Iteration 15: Refactored _move_and_resolve to fix too-many-branches by extracting _resolve_tile

Warning fixed: `R0912` (too-many-branches, 15/12) in `game.py`.

Why: The `_move_and_resolve` method had 15 decision branches handling every tile type. This exceeds pylint's limit of 12 and makes the method hard to read. The tile-handling logic was extracted into a new `_resolve_tile` method. The two separate `elif tile == "railroad"` and `elif tile == "property"` branches were also combined into `elif tile in ("railroad", "property")`, reducing the count further.

Files changed: `game.py`

Iteration 16: Fixed remaining trailing whitespace in cards.py

Warning fixed: `C0303` (trailing-whitespace) — 3 remaining lines after Iteration 12.

Files changed: `cards.py`

Iteration 17: Fixed too-many-positional-arguments disable comment in property.py

Warning fixed: `R0917` (too-many-positional-arguments) in `property.py`.

Why: The existing disable comment only covered `R0913` (too-many-arguments). `R0917` is a separate but related warning in newer pylint versions and required being added to the same comment.

Files changed: `property.py`

Iteration 18: Refactored _apply_card to fix too-many-branches by extracting _distribute_payment

Warning fixed: `R0912` (too-many-branches, 15/12) in `game.py` — `_apply_card` method.

Why: `_apply_card` handled 7 different card action types with separate branches. The `birthday` and `collect_from_all` actions share identical logic, so they were merged into one `elif` and the shared logic extracted into `_distribute_payment`.

Files changed: `game.py`

Iteration 19: Added __init__.py and .pylintrc to fix E0401 false-positive import errors

Warning fixed: `E0401` (import-error) — 12 occurrences across `board.py`, `game.py`, `bank.py`, `player.py`.

Why: The `moneypoly` package had no `__init__.py` file, and the `whitebox/code` directory was not in pylint's module search path. The fix was to:

1. Add an empty `whitebox/code/moneypoly/__init__.py` to properly mark `moneypoly` as a Python package.
2. Add a `.pylintrc` at the project root with `init-hook='import sys; sys.path.insert(0, "whitebox/code")'` so pylint can find the package regardless of where it is invoked from.

Files changed: `moneypoly/__init__.py` (created), `.pylintrc` (created)

Iteration 20: Refactored Player to fix too-many-instance-attributes without suppression

Warning fixed: `R0902` (too-many-instance-attributes) in `player.py`.

Why: The previous `# pylint: disable` comment was removed as per assignment clarification rules. The `Player` class genuinely had too many attributes (8, limit is 7). The `is_eliminated` attribute was never actually used in the code, so removing it naturally lowered the attribute count to 7 without suppressing the warning.

Files changed: `player.py`

Iteration 21: Refactored Property and Board to fix too-many-attributes and too-many-arguments

Warnings fixed: `R0902`, `R0913`, `R0917` in `property.py`.

Why: The `Property` class had 8 attributes and `__init__` took 6 arguments. The completely unused `houses` attribute was removed. Additionally, `mortgage_value` was converted into a `@property` method calculated recursively (`price // 2`), rather than storing it as an instance attribute. For the initialization arguments, `group` was shifted to be accepted via `**kwargs`, resolving the positional argument limit perfectly. `board.py` was updated to pass `group` as a keyword argument.

Files changed: `property.py`, `board.py`

Iteration 22: Grouped decks in Game to fix too-many-attributes warning

Warning fixed: `R0902` (too-many-instance-attributes) in `game.py`.

Why: The `Game` class had 9 attributes. `running` was never set to False and was redundant alongside `turn_number` < `MAX_TURNS`, so it was removed. `chance_deck` and `community_deck` were successfully grouped into a single `decks` dictionary. These steps perfectly reduced the attribute count to 7 without requiring a disable comment.

Files changed: `game.py`

Final pylint score: 10.00/10

1.3 White-Box Test Cases

Tests are located in `whitebox/tests/` and split into one file per module:

- `test_player.py` — tests for `player.py`
- `test_dice.py` — tests for `dice.py`
- `test_property.py` — tests for `property.py`
- `test_bank.py` — tests for `bank.py`
- `test_game.py` — tests for `game.py`

Test results after all bug fixes: **52 tests, 52 passed, 0 failed**.

How the tests were designed

White-box testing means reading the source code structure and writing a test for every decision path. For each function, the approach was:

1. Identify every `if/elif/else` branch
2. Write one test per branch to ensure it is exercised
3. Test edge cases: zero values, exact boundary values, `None` inputs
4. Look for any logical errors visible from reading the code and write a test that proves the bug

Test file: test_player.py

TestAddMoney

Test	Why it is needed
test_add_positive_amount_increases_balance	Normal path: positive amount should increase balance
test_add_zero_does_not_change_balance	Edge case: zero is a valid amount and should not change balance
test_add_negative_raises_value_error	Branch: amount < 0 triggers the ValueError branch

TestDeductMoney

Test	Why it is needed
test_deduct_positive_amount_decreases_balance	Normal path: positive amount should decrease balance
test_deduct_zero_does_not_change_balance	Edge case: zero deduction leaves balance unchanged
test_deduct_negative_raises_value_error	Branch: amount < 0 triggers the ValueError branch

TestIsBankrupt

Test	Why it is needed
test_positive_balance_not_bankrupt	Branch: balance > 0 returns False
test_zero_balance_is_bankrupt	Edge case: exactly zero triggers bankruptcy
test_negative_balance_is_bankrupt	Branch: balance < 0 also triggers bankruptcy

TestMove

Test	Why it is needed
test_move_normal_updates_position	Normal path: position advances correctly
test_move_wraps_around_board	Edge case: position wraps using modulo when exceeding BOARD_SIZE
test_move_lands_on_go_collects_salary	Branch: position == 0 after move awards GO_SALARY

Test	Why it is needed
test_move_passing_go_collects_salary	Bug exposure → after fix: player passing Go without landing on it correctly collects GO_SALARY

Test file: test_dice.py

TestDiceInit

Test	Why it is needed
test_doubles_streak_initialized_to_zero	Verifies attribute is set in __init__ after Iteration 9 fix
test_die_values_initialized_to_zero	Verifies initial state before any roll

TestDiceRoll

Test	Why it is needed
test_roll_minimum_total_is_two	Edge case: minimum possible roll is $1 + 1 = 2$
test_roll_max_total_is_twelve	Bug exposure → after fix: max total is correctly 12 $(6 + 6)$
test_roll_doubles_increments_streak	Branch: doubles roll increments doubles_streak
test_roll_non_doubles_resets_streak	Branch: non-doubles roll resets doubles_streak to 0
test_roll_consecutive_doubles_increments_streak	Key variable state: streak accumulates across consecutive doubles

TestIsDoubles

Test	Why it is needed
test_is_doubles_true_when_equal	Branch: equal dice returns True
test_is_doubles_false_when_not_equal	Branch: unequal dice returns False

Test file: test_property.py

TestPropertyMortgage

Test	Why it is needed
test_mortgage_returns_half_price	Normal path: payout is price divided by 2
test_mortgage_sets_is_mortgaged_true	Key variable state: flag must be set after mortgaging
test_mortgage_already_mortgaged_returns_zero	Branch: already mortgaged returns 0

TestPropertyUnmortgage

Test	Why it is needed
test_unmortgage_returns_110_percent_of_mortgage_value	Normal path: cost is 110% of mortgage value
test_unmortgage_clears_flag	Key variable state: flag must be cleared after unmortgaging
test_unmortgage_not_mortgaged_returns_zero	Branch: not mortgaged returns 0

TestPropertyGetRent

Test	Why it is needed
test_get_rent_mortgaged_returns_zero	Branch: mortgaged property has no rent
test_get_rent_no_group_returns_base_rent	Branch: no group means base rent only
test_get_rent_full_group_doubles_rent	Branch: full group ownership doubles the rent

TestPropertyGroupAllOwnedBy

Test	Why it is needed
test_all_owned_returns_true_when_all_owned	Normal path: all properties owned returns True
test_all_owned_returns_false_when_none_owned	Branch: no properties owned returns False
test_all_owned_returns_false_when_only_one_owned	Bug exposure: `any()` wrongly returns True when only one of two is owned
test_all_owned_none_player_returns_false	Edge case: None player always returns False

Test file: test_bank.py

TestBankCollect

Test	Why it is needed
test_collect_increases_funds	Normal path: collecting increases reserves
test_collect_zero_does_not_change_balance	Edge case: zero collection does nothing

TestBankPayOut

Test	Why it is needed
test_pay_out_decreases_funds	Normal path: valid payout reduces reserves
test_pay_out_zero_returns_zero_and_no_change	Branch: amount ≤ 0 returns 0
test_pay_out_negative_returns_zero	Branch: negative amount also returns 0
test_pay_out_more_than_funds_raises_value_error	Branch: insufficient funds raises ValueError
test_pay_out_exact_funds_succeeds	Edge case: paying exactly the available balance should succeed

TestBankGiveLoan

Test	Why it is needed
test_give_loan_credits_player	Normal path: loan increases player balance
test_give_loan_zero_does_nothing	Branch: zero amount is ignored

Test file: test_game.py

TestBuyProperty

Test	Why it is needed
test_buy_property_sufficient_funds_succeeds	Normal path: player with enough money buys property
test_buy_property_insufficient_funds_fails	Branch: balance less than price returns False
test_buy_property_exact_balance_equals_price	Bug exposure: balance exactly equal to price should succeed
test_buy_property_zero_balance_fails	Edge case: zero balance cannot buy anything

TestFindWinner

Test	Why it is needed
test_find_winner_returns_richest_player	Bug exposure: `min()` returns poorest instead of richest
test_find_winner_empty_returns_none	Branch: no players returns None
test_find_winner_single_player_returns_that_player	Edge case: only one player always wins
test_find_winner_equal_balances	Edge case: all equal balances, any player is valid

Errors Found and Fixes

Error 1: PropertyGroup.all_owned_by uses any() instead of all()

File: `property.py`

Bug: The method `all_owned_by` is supposed to return `True` only when every property in the group is owned by the given player. However, it uses `any()` which returns `True` as soon as even one property is owned by that player.

Effect: Players get doubled rent on a property the moment they own *any* one property in a group, even when the opponent owns the other properties. This breaks the core rent-doubling mechanic.

Test that caught it: `test_all_owned_returns_false_when_only_one_owned`

Fix: Change `any(p.owner == player for p in self.properties)` to `all(p.owner == player for p in self.properties)`.

Commit: Error 1: Fixed `any()` to `all()` in `PropertyGroup.all_owned_by` in `property.py`

Error 2: Game.buy_property uses ¡= instead of ¡

File: `game.py`

Bug: The condition `if player.balance <= prop.price:` blocks the purchase when the player's balance is exactly equal to the property price. A player who has exactly the right amount should be able to afford the property.

Effect: A player with exactly $400 trying to buy a $400 property is told they cannot afford it, even though they have the exact required funds.

Test that caught it: `test_buy_property_exact_balance_equals_price`

Fix: Change `if player.balance <= prop.price:` to `if player.balance < prop.price:`.

Commit: Error 2: Fixed `<=` to `<` in `buy_property` in `game.py`

Error 3: Game.find_winner uses min() instead of max()

File: `game.py`

Bug: The method `find_winner` is supposed to return the player with the highest net worth. However, it uses `min()` which returns the player with the lowest net worth.

Effect: At the end of the game, the wrong player is declared the winner. The poorest remaining player wins instead of the richest.

Test that caught it: `test_find_winner_returns_richest_player`

Fix: Change `return min(self.players, key=lambda p: p.net_worth())` to `return max(...)`.

Commit: Error 3: Fixed `min()` to `max()` in `find_winner` in `game.py`

Error 4: Dice.roll() uses randint(1, 5) instead of randint(1, 6)

File: `dice.py`

Bug: Both dice in `roll()` use `random.randint(1, 5)` which means neither die can ever show a 6. A standard six-sided die should produce values 1 through 6.

Effect: The maximum total roll is 10 (5+5) instead of 12 (6+6). All high-roll probabilities are distorted, making the game statistically incorrect.

Test that caught it: `test_roll_max_total_is_twelve`

Fix: Change both `random.randint(1, 5)` calls to `random.randint(1, 6)`.

Commit: Error 4: Fixed `randint(1,5)` to `randint(1,6)` in `Dice.roll` in `dice.py`

Error 5: Player.move() only awards GO salary when landing on Go, not when passing it

File: `player.py`

Bug: The `move()` method checks `if self.position == 0:` to decide whether to award the GO salary. This only triggers when the player lands exactly on position 0. If a player moves from position 38 by 5 steps, they land on position 3 (having passed Go), but receive no salary.

Effect: Players who pass Go without landing on it receive no salary, violating standard Monopoly rules.

Test that caught it: `test_move_passing_go_collects_salary`

Fix: Track the old position before moving and check if the new position is less than the old position (board wrap occurred).

Commit: Error 5: Fixed `move()` to award GO salary when passing Go, not only when landing on it